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CS 5365 - Deep Learning: Final Project Report

GitHub Link: https://github.com/djbooth-3/DeepLearningProject

Evaluating MAMBA on Continuous Sensor Data

Introduction

Sequence modeling is fundamental to many machine learning tasks, yet the dominant Transformer architecture suffers from quadratic complexity in sequence length due to its self-attention mechanism. This becomes a bottleneck when processing long sequences such as time-series and audio. Structured State Space Models (SSMs) have emerged as a promising alternative, offering linear-time computation while maintaining strong sequence modeling capability. Our project evaluates the performance of Mamba: Linear-Time Sequence Modeling with Selective State Spaces by Albert Gu and Tri Dao (COLM 2024) [1]. The paper's primary contribution is a selection mechanism that makes SSM parameters input-dependent, allowing the model to selectively retain or discard information at each timestep addressing the core limitation of prior SSMs, which were linear time-invariant (LTI) and could not adapt their dynamics based on content. We evaluate Mamba using the WESAD physiological stress detection dataset, testing whether its scalability advantage over Transformers, LSTMs, and CNNs holds on long continuous sensor signals, a domain not covered in the original paper.

Chosen Result

We aimed to reproduce the paper's claim that Mamba scales more effectively than Transformers on longer sequences. We expect Mamba to maintain or improve classification accuracy as window size increases from 60 seconds to 120 seconds, while the Transformer model degrades due to its attention cost. This result was chosen because scalability is Mamba's core design motivation. Reproducing it on real-world physiological sensor data, a continuous signal which is not studied in the original paper directly, tests whether the advantage generalizes beyond language.

Methodology

We use the Wearable Stress and Affect Detection (WESAD) dataset [2]. This includes 15 subjects, 8 physiological channels (ACC (3), ECG, EDA, EMG, Resp, Temp), and 3 affective states: baseline, stress, and amusement. Signals were recorded at 700Hz from a chest worn device. For prepossessing, we downsampled from 700 Hz to 100 Hz to allow for faster training. Additionally, we segmented the sensor data into 60, 90, and 120 second sliding windows with 30 second stride. We performed a subject-aware train/test split, subjects were assigned entirely to either train or test, preventing any individual's data from appearing in both sets. The data was split by 70% train, 15% validation, and 15% test. Class-balanced loss was also applied to account for the unequal distribution of affective states across windows.

Four architectures were trained and evaluated under identical conditions on the same data splits:

Mamba: A selective SSM that makes its parameters input-dependent, allowing the model to selectively propagate or forget information based on context.

Transformer: A self-attention mechanism to weigh the importance of every part of the input sequence relative to every other part.

LSTM: A specialized recurrent neural network (RNN) that uses cell state and three internal gates (input, forget, and output) to regulate what information is kept or discarded. Additionally, we tested a bi-directional version, which processes data in both forward and backward directions. This allows the model to gain access to both past and future contexts.

CNN: Uses convolutional layers to scan small local regions of data to extract hierarchical features like edges, shapes, and complex patterns.

Each architecture was evaluated by the overall test accuracy and F1 score, with a particular focus on stress detection F1 as the most relevant class.

Results & Analysis

Model	**60s Acc**	**60s Stress F1**	**90s Acc**	**90s Stress F1**	**120s Acc**	**120s Stress F1**
Mamba	**72.9%**	**0.61**	**76.6%**	**0.67**	**68.3%**	**0.53**
Transformer	**61.4%**	**0.34**	**41.9%**	**0.31**	**53.3%**	**0.55**
Bi-LSTM	**65.3%**	**0.43**	**59.5%**	**0.36**	**36.7%**	**0.38**
CNN	**72.9%**	**0.74**	**72.9%**	**0.74**	**74.9%**	**0.72**

Our results support the paper's scalability claim. Mamba was the most consistent model across all window sizes. It consistently outperformed the Transformer and LSTM, achieving its best overall accuracy at 90s. Mamba never collapsed under longer sequences, maintaining stable accuracy and stress F1 from 60s through 120s. Longer context benefited Mamba supporting the value of selective state spaces for signals. The Transformer's degradation was severe: accuracy dropped from 61.4% at 60s to 41.9% at 90s then 53.3% at 120%. LSTM also fell behind confirming its known vulnerability to very long sequences.

CNN was a surprisingly strong competitor, maintaining the highest stress F1 across all window sizes, suggesting local temporal patterns alone are highly informative for stress detection. Overall, Mamba offers the best balance of accuracy, stability, and scalability, though CNN's consistent stress precision remains a compelling result worth further investigation.

Reflections

The most important lesson from this project was that Mamba's theoretical advantages can be seen even at modest scales. We did not need million-token sequences to see the Transformer's decline or Mamba's stable scaling. The CNN result challenged our initial assumptions significantly. We expected CNN to perform worse given its lack of global context modeling. Instead, it raised a genuine research question: for tasks where the signal of interest is local in time, does a global sequence model actually help? We also learned the importance of subject-aware splitting. Future directions include Deep Mamba architecture, Bidirectional Mamba architecture, Evaluating performance under simulated sensor dropout, and Subject covariate integration.

References

1. Gu, Albert, and Tri Dao. "Mamba: Linear-time sequence modeling with selective state spaces." arXivpreprint arXiv:2312.00752 (2023).
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3. State Spaces. (2023). *Mamba Official Repository*. https://github.com/state-spaces/mamba